

Julio Cesar Garcia

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EDUCATION - The University of Texas at El Paso (UTEP) - Cumulative GPA: 3.75/4.00

Bachelor of Science in Mechanical Engineering - With a minor in Mathematics
Master of Science in Mechanical Engineering

Graduation Date: 05/2023

Anticipated Date: 06/2026

Honors, Awards & Affiliations:

College of Engineering Dean's List

Fall 19, Spring 21-Fall 22, Spring 24-Present

3D Modeling of Space Systems and Concepts, Member

07/2021-07/2024

EXPERIENCE

City of El Paso - Engineering Senior Technician

7/2025 - Present

- Review, analyze, and approve **Traffic Control Plans** to ensure compliance with **MUTCD** standards, City specifications, and work zone safety requirements.
- Review and approve **Pavement Cut Permits**, evaluating cutting limits, materials, restoration methods, and construction sequencing for asphalt and concrete streets.
- Verify proper placement, spacing, visibility, and sequencing of **traffic control devices**, including lane closures, detours, flagging, and pedestrian accommodations.
- Validate field execution of approved TCPs and pavement restorations alongside City inspectors to ensure real-world compliance and safety.
- Serve as a technical liaison between contractors, inspectors, engineers, and City departments to resolve plan conflicts and maintain permit compliance.

Space Exploration Technology Research Center - Manufacturing Graduate Research - UTEP 9/2021-12/2023, 1/2025 - 3/2025

Project advisor: Dr. Flores - afloresabad@utep.edu

- Designed and manufactured a **CNC-compatible 3-DOF** robotic arm for U2 satellite integration, generated **G-code** using NX12 and Fusion 360 CAM.
- Digitized** physical components and led their production using **CNC machining**, **lathe**, **mill**, and advanced manufacturing methods.
- Led the **design** and **manufacturing** of a high-pressure interface device rated to 3000 psi, enabling precise control of airflow for one-plane space testing.
- Engineered and built an **air-bearing sled** simulating space conditions, improving test accuracy by 33%.
- Mentored 4 interns and collaborated with multidisciplinary teams to meet research and testing objectives.

Hiller Measurements Innovative Solutions - Proteus ATE – Engineer Intern

12/2023-10/2024

- Pioneered the **design** of a high-performance, **manufacturable** rack door, incorporating an Ergotron arm for seamless technician access, significantly enhancing **operational efficiency**.
- Enhanced Proteus ATE software by refining a 3D page viewer as well as developing the cable addressing **logic**, demonstrating adaptability and problem-solving skills in **coding**.

PROJECTS

BAJA Society of Automotive Engineers (SAE) - Steering and Brakes team leader

02/2022-02/2023

- Led a 10-member steering and brakes team, training members in **CAD**, **FEA**, **CAM**, **CNC machining**, and automotive fundamentals.
- Coordinated with **cross-functional sub-teams** to identify bottlenecks, improve workflows, and meet competition deadlines.
- Directed the **design**, **fabrication**, and **testing** of steering components, utilizing both **CNC machinery** and conventional methods to achieve the **highest precision**. This hands-on approach ensured reliability, safety, and optimal performance.

Manufacturing Engineering Lab, Mechanical Spider

08/2022-12/2022

- Team lead - **Designed** a functioning spider that controls 8 different legs with only 1 actuator. **Managed** a group of 12 people by assisting and allocating different tasks needed to complete the project. Fully **manufactured** spider out of aluminum plates.
- Conducted comprehensive research into the materials, components, costs, and methodologies required for the spider.

SKILLS

- Software:** Autodesk Fusion 360, Autodesk 3D Inventor, Autodesk Tinkercad, Autodesk Tinkercad Circuits, AutoCAD, Blender, NX 12.0, Bambu Studio, Ultimaker Cura 4.13.0, Creality Slicer, Python, GitHub, HTML, CSS, Arduino IDE 1.8.19, MATLAB R2020R, NI LabVIEW 2022, Simulink, Unity, Unreal Engine 5.0.1.
- Microsoft Office Certified by Certiport:** Word 2016, Excel 2016, Outlook 2016, and PowerPoint 2016.
- Metal tool knowledge of a metal machine shop. Welding experience and teamwork. Switchboard knowledge.
- Drawing/Schematic/Blueprint/Plan Reading.
- Bilingual:** English and Spanish.